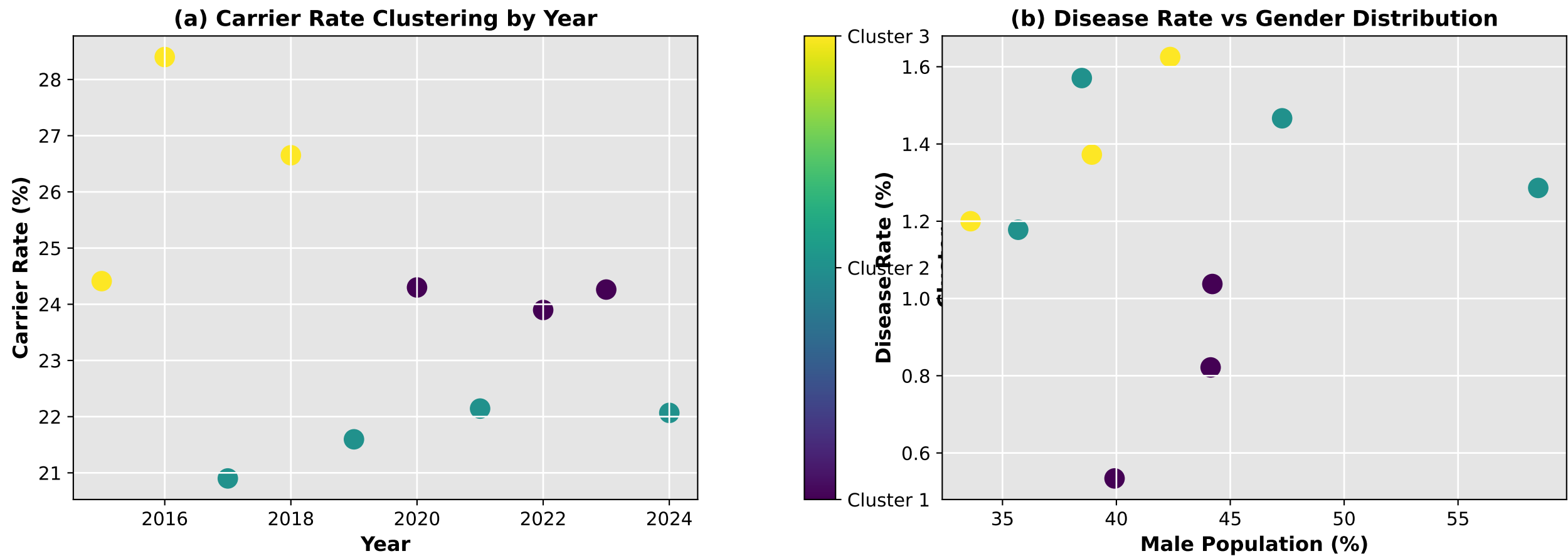


Fig. 5. Clustering Analysis of Genotype Distribution Patterns



Clustering Results Summary:

Cluster 1 (Stabilisation): Years represented: [2020, 2022, 2023]
Avg Carrier Rate: 24.15%
Avg Disease Rate: 0.80%
Avg Male Rate: 42.76%

Cluster 2 (Transition): Years represented: [2017, 2019, 2021, 2024]
Avg Carrier Rate: 21.68%
Avg Disease Rate: 1.38%
Avg Male Rate: 44.99%

Cluster 3 (High-burden): Years represented: [2015, 2016, 2018]
Avg Carrier Rate: 26.49%
Avg Disease Rate: 1.40%
Avg Male Rate: 38.29%